



KNIME

Data Analytics &
Data Engineering

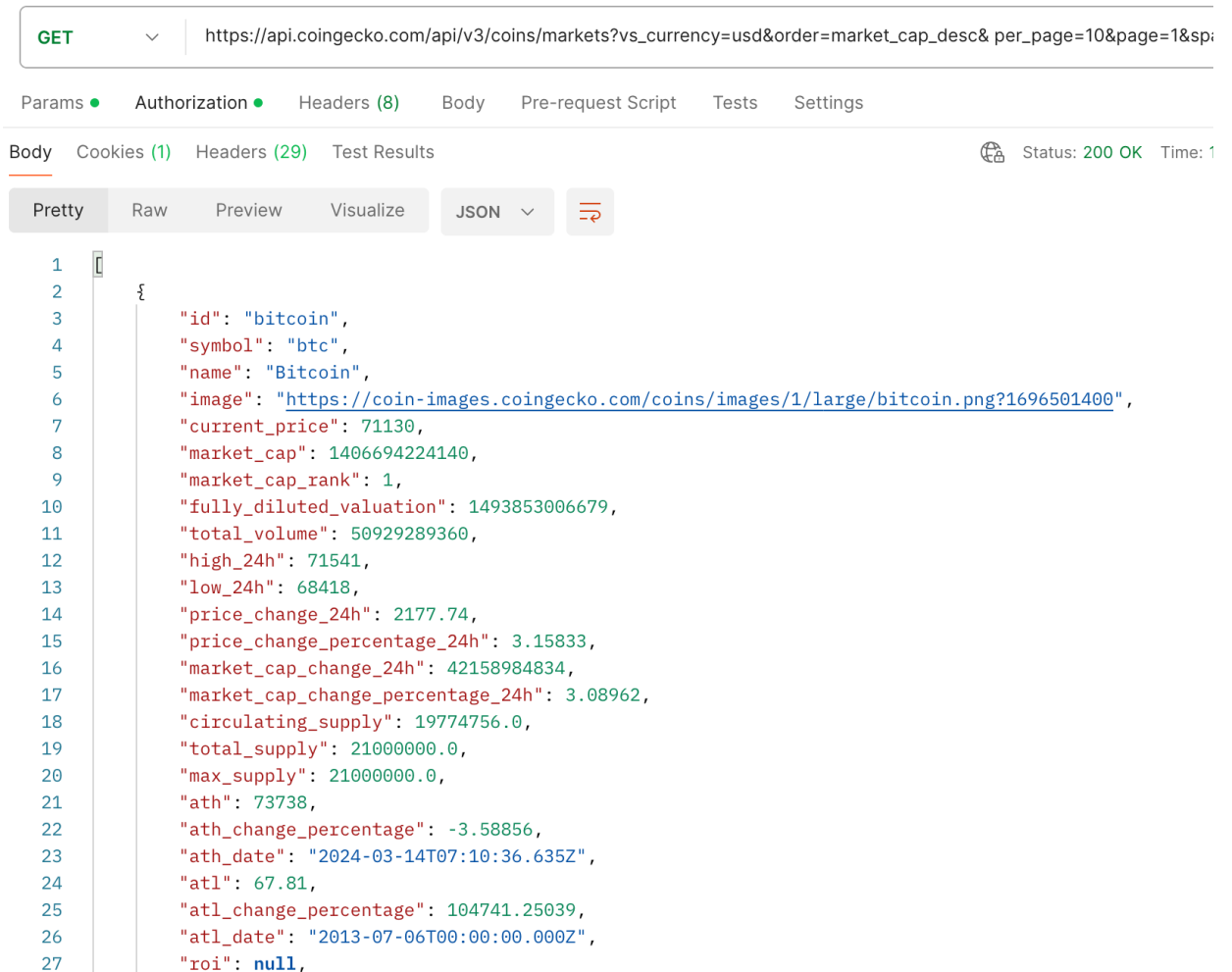
Application Programming Interface: API

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REST API

- ทดสอบเรียก REST API ด้วย POSTMAN

https://api.coingecko.com/api/v3/coins/markets?vs_currency=usd&order=market_cap_desc&per_page=10&page=1&sparkline=false&price_change_percentage=7d



The screenshot shows a Postman interface with a GET request to the CoinGecko API. The URL is `https://api.coingecko.com/api/v3/coins/markets?vs_currency=usd&order=market_cap_desc&per_page=10&page=1&sparkline=false&price_change_percentage=7d`. The response is a JSON object for Bitcoin, displaying various market metrics.

```
1 {
2   "id": "bitcoin",
3   "symbol": "btc",
4   "name": "Bitcoin",
5   "image": "https://coin-images.coingecko.com/coins/images/1/large/bitcoin.png?1696501400",
6   "current_price": 71130,
7   "market_cap": 1406694224140,
8   "market_cap_rank": 1,
9   "fully_diluted_valuation": 1493853006679,
10  "total_volume": 50929289360,
11  "high_24h": 71541,
12  "low_24h": 68418,
13  "price_change_24h": 2177.74,
14  "price_change_percentage_24h": 3.15833,
15  "market_cap_change_24h": 42158984834,
16  "market_cap_change_percentage_24h": 3.08962,
17  "circulating_supply": 19774756.0,
18  "total_supply": 21000000.0,
19  "max_supply": 21000000.0,
20  "ath": 73738,
21  "ath_change_percentage": -3.58856,
22  "ath_date": "2024-03-14T07:10:36.635Z",
23  "atl": 67.81,
24  "atl_change_percentage": 104741.25039,
25  "atl_date": "2013-07-06T00:00:00.000Z",
26  "roi": null,
27 }
```

REST API

- ทดสอบเรียก REST API ด้วย KNIME

[https://api.coingecko.com/api/v3/coins/markets?vs_currency=usd&order=market_cap_desc&](https://api.coingecko.com/api/v3/coins/markets?vs_currency=usd&order=market_cap_desc&per_page=10&page=1&sparkline=false&price_change_percentage=7d)

[per_page=10&page=1&sparkline=false&price_change_percentage=7d](https://api.coingecko.com/api/v3/coins/markets?vs_currency=usd&order=market_cap_desc&per_page=10&page=1&sparkline=false&price_change_percentage=7d)

- จากนั้นทำการ Filter เอาเฉพาะเหรียญที่มี market_cap_rank 1-5

GET Request



GET Request

Dialog - 5:1 - GET Request

Connection Authentication Proxy Error Handling Request Headers Response Headers Flow Variables Job Manager Selection Memory Policy

URL: https://api.coingecko.com/api/v3/coins/markets?vs_currency=usd&order=market_cap_desc&per_page=10&page=1&sparkline=false&price_change_percentage=7d

URL column:

Handling of invalid URLs

☒ Insert missing value ☐ Fail node execution immediately ☐ Omit row from output

☐ Delay (ms): 0

Concurrency: 1

SSL

☐ Ignore hostname mismatches ☐ Trust all certificates

☒ Follow redirects

☒ Send large data in chunks

Timeout (s): 2

Body column: body

OK Apply Cancel ?

GET Request

GET results (Table)

Rows: 1

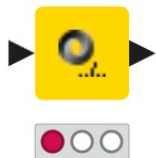
Columns: 3

#	RowID	Status	Content type	body
		Number (integer)	String	JSON
1	Row0	200	application/json; charset=utf-8	<pre>{ "id": "bitcoin", "symbol": "btc", "name": "Bitcoin", "image": "https://coin-images.coingecko.com...", "current_price": 68445, "market_cap": 1353894327828, "market_cap_rank": 1, "fully_diluted_valuation": 1438152534591, "total_volume": 36219203522, "high_24h": 68970, "low_24h": 67526, "price_change_24h": 918.93, "price_change_percentage_24h": 1.36087, "market_cap_change_24h": 18712389720, "market_cap_change_percentage_24h": 1.401..., "circulating_supply": 19769656, "total_supply": 21000000, "max_supply": 21000000, "ath": 73738, "ath_change_percentage": -7.17238, "ath_date": "2024-03-14T07:10:36.635Z", "atl": 67.81, "atl_change_percentage": 100844.07992, "atl_date": "2013-07-06T00:00:00.000Z", "roi": null, "last_updated": "2024-10-19T03:59:38.437Z", "price_change_percentage_7d_in_currency": ... }, { "id": "ethereum", "symbol": "eth", "name": "Ethereum", "image": "https://coin-images.coingecko.com...", "current_price": 2646.49, "market_cap": 318630708556, "market_cap_rank": 2, "fully_diluted_valuation": 318630708556, "total_volume": 14324376946, "high_24h": 2671.32, "low_24h": 2609.35, "price_change_24h": 37.14, "price_change_percentage_24h": 1.42334 }</pre>

- หลังจาก Execute จะได้ Status, Content-Type, และ Body (JSON Arrays)

JSON Path

JSON Path



Dialog - 5:2 - JSON Path

Settings | Flow Variables | Job Manager Selection | Memory Policy

Input
(JSON) body

☐ Remove source column

Outputs

Output column	JSONPath	List	Paths
D atl	\$[*]['atl']	☒	☐
D atl_change_percentages	\$[*]['atl_change_percentage']	☒	☐
S atl_dates	\$[*]['atl_date']	☒	☐
S last_updateds	\$[*]['last_updated']	☒	☐
D price_change_percentage_7d_in_currencys	\$[*]['price_change_percentage_7d_in_currency']	☒	☐
D times	\$[*]['roi']['times']	☒	☐
S currencys	\$[*]['roi']['currency']	☒	☐
D percentages	\$[*]['roi']['percentage']	☒	☐

Add single query Add collection query Add JSONPath Edit JSONPath Remove JSONPath ↑ ↓

JSON-Cell Preview

```
1 [ {  
2   "id" : "bitcoin",  
3   "symbol" : "btc",  
4   "name" : "Bitcoin",  
5   "image" : "https://coin-images.coingecko.com/coins/images/1/large/bitcoin.png?1696501400",  
6   "current_price" : 68445,  
7   "market_cap" : 1353894327828,  
8   "market_cap_rank" : 1,  
9   "fully_diluted_valuation" : 1438152534591,  
10  "total_volume" : 36219203522,  
11  "high_24h" : 68970.
```

OK Apply Cancel ?

JSON Path

เลือก cell ที่ต้องการ

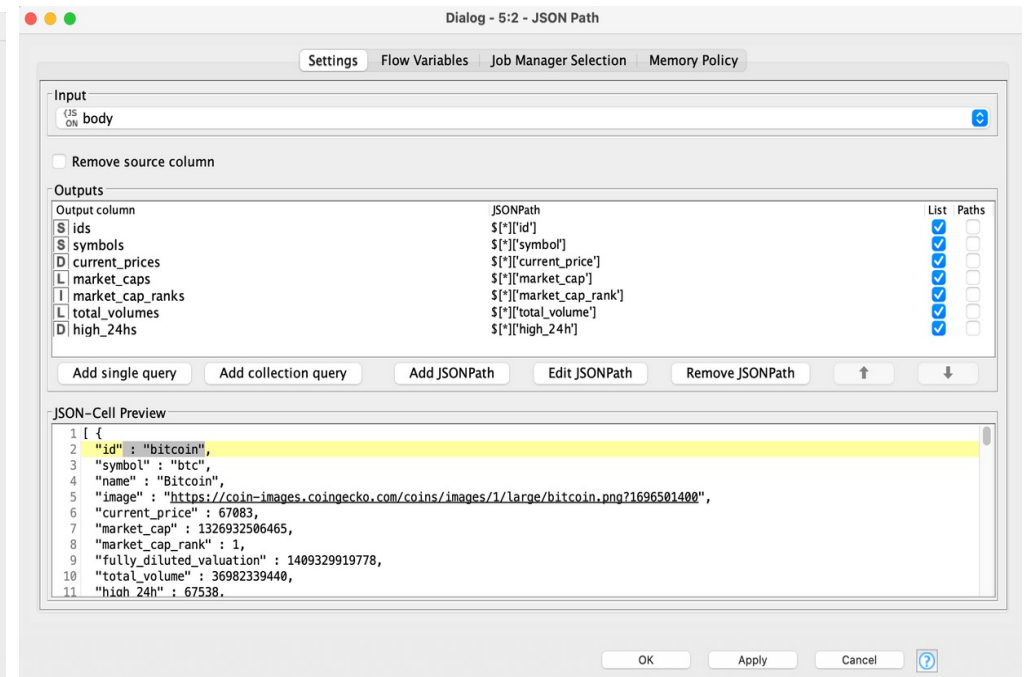
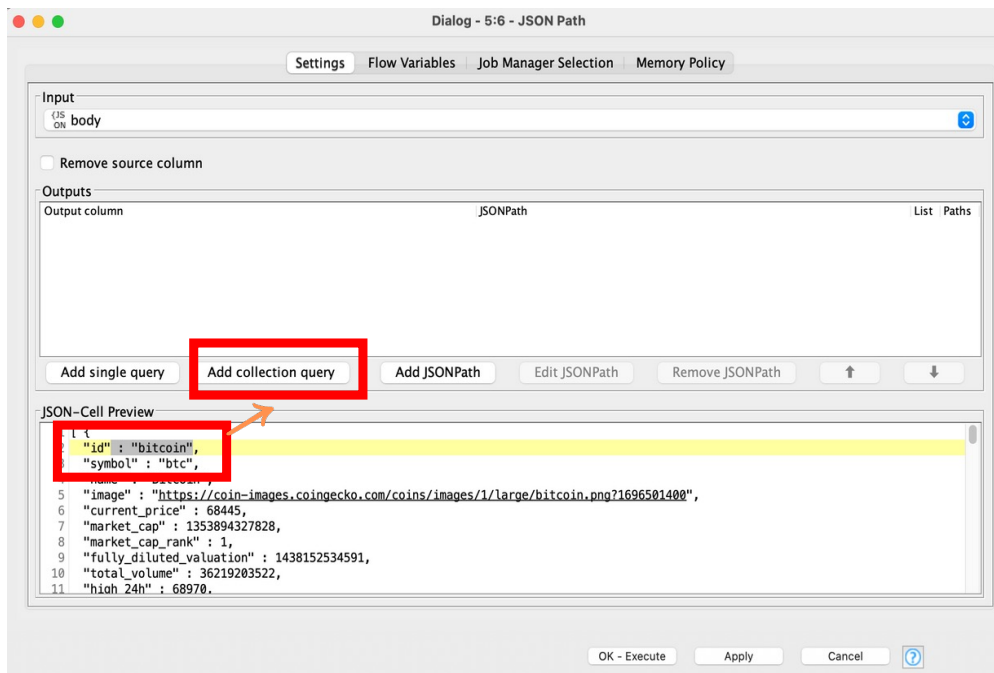


Table with 11 columns: #, RowID, S..., C..., b..., ids, symbols, names, images, current_prices, market_caps, market_c... The table shows data for Bitcoin, including its ID, symbol, name, image URL, current price, market cap, and other metrics. An orange arrow points to the 'market_c...' column header.

#	RowID	S...	C...	b...	ids	symbols	names	images	current_prices	market_caps	market_c...
1	Row0	200	appli...	[	[bitcoin,ethereum,tether,...]	[btc,eth,usdt,...]	[Bitcoin,Ethereum,Tether,...]	[https://c...	[68445.0,2646.49,0.999584,...]	[1353894327828,318630708556,119981028809,...]	[1,2,3,...]

UNGROUP

Rows: 1 | Columns: 12

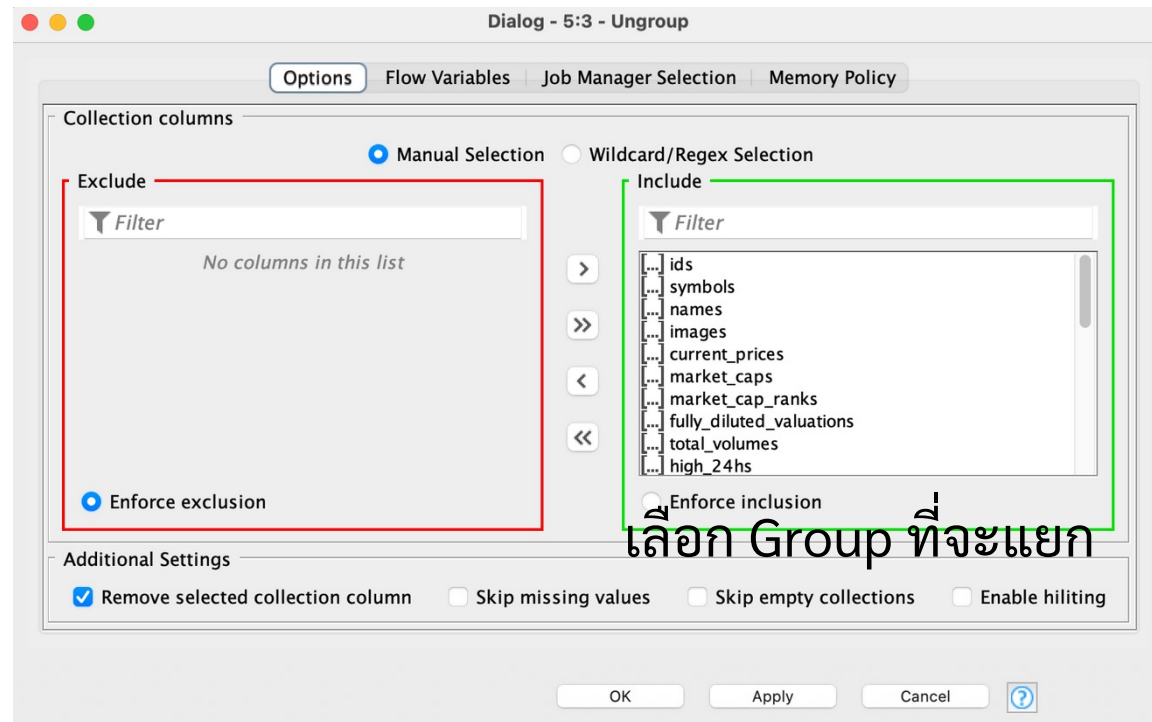
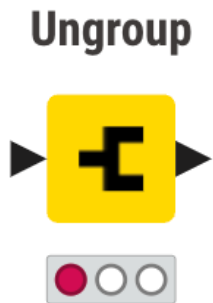
#	RowID	S...	C...	b...	ids	symbols	names	images	current_prices	market_caps	market_c...
1	Row0	200	appli...	[	[bitcoin,ethereum,tether,...]	[btc,eth,usdt,...]	[Bitcoin,Ethereum,Tether,...]	[https://c...	[68445.0,2646.49,0.999584,...]	[1353894327828,318630708556,119981028809,...]	[1,2,3,...]



Rows: 10 | Columns: 32

#	RowID	S...	C...	b...	ids	symbols	names	images	current_p...	market_c...	market_c...	fully_dilut...	total_volu...	high_24hs	low_24hs	price_cha...
1	Row...	200	appli...	[	bitcoin	btc	Bitcoin	https://coin-i...	68,445	1353894327...	1	1438152534...	36219203522	68,970	67,526	918.93
2	Row...	200	appli...	[	ethereum	eth	Ethereum	https://coin-i...	2,646.49	318630708556	2	318630708556	14324376946	2,671.32	2,609.35	37.14
3	Row...	200	appli...	[	tether	usdt	Tether	https://coin-i...	1	119981028809	3	119981028809	57529392091	1.004	0.997	0.001
4	Row...	200	appli...	[	binancecoin	bnb	BNB	https://coin-i...	598.83	87399219145	4	87399219145	550370994	602.55	593.47	5.36
5	Row...	200	appli...	[	solana	sol	Solana	https://coin-i...	154.62	72676364138	5	90757778069	1938160138	156.46	151.76	2.86
6	Row...	200	appli...	[	usd-coin	usdc	USDC	https://coin-i...	1	34937218632	6	34942311895	4575656070	1.003	0.997	0
7	Row...	200	appli...	[	ripple	xrp	XRP	https://coin-i...	0.548	31051653136	7	54751548989	1091918006	0.552	0.542	0.004
8	Row...	200	appli...	[	staked-ether	steth	Lido Staked E...	https://coin-i...	2,646.29	25850084860	8	25850084860	30168489	2,670.65	2,608.92	37.37
9	Row...	200	appli...	[	dogecoin	doge	Dogecoin	https://coin-i...	0.145	21247031142	9	21250176012	2409221626	0.146	0.132	0.012
10	Row...	200	appli...	[	tron	trx	TRON	https://coin-i...	0.158	13708756876	10	13708773705	294328153	0.16	0.158	-0

UNGROUP



Column Filter

Rows: 10 | Columns: 32

☐	#	RowID	S... N...	C... St...	b... J...	ids String	symbols String	names String	images String	current_p... Number (dou...	market_c... Number (long)	market_c... Number (inte...	fully_dilut... Number (long)	total_volu... Number (long)	high_24hs Number (dou...	low_24hs Number (dou...	price_cha... Number (dou...	
☐	1	Row...	200	appli...	[	bitcoin	btc	Bitcoin	https://coin-i...	68,445	1353894327...	1	1438152534...	36219203522	68,970	67,526	918.93	1.
☐	2	Row...	200	appli...	[	ethereum	eth	Ethereum	https://coin-i...	2,646.49	318630708556	2	318630708556	14324376946	2,671.32	2,609.35	37.14	1.
☐	3	Row...	200	appli...	[	tether	usdt	Tether	https://coin-i...	1	119981028809	3	119981028809	57529392091	1.004	0.997	0.001	0.
☐	4	Row...	200	appli...	[	binancecoin	bnb	BNB	https://coin-i...	598.83	87399219145	4	87399219145	550370994	602.55	593.47	5.36	0.
☐	5	Row...	200	appli...	[	solana	sol	Solana	https://coin-i...	154.62	72676364138	5	90757778069	1938160138	156.46	151.76	2.86	1.
☐	6	Row...	200	appli...	[	usd-coin	usdc	USDC	https://coin-i...	1	34937218632	6	34942311895	4575656070	1.003	0.997	0	0.
☐	7	Row...	200	appli...	[	ripple	xrp	XRP	https://coin-i...	0.548	31051653136	7	54751548989	1091918006	0.552	0.542	0.004	0.
☐	8	Row...	200	appli...	[	staked-ether	steth	Lido Staked E...	https://coin-i...	2,646.29	25850084860	8	25850084860	30168489	2,670.65	2,608.92	37.37	1.
☐	9	Row...	200	appli...	[	dogecoin	doge	Dogecoin	https://coin-i...	0.145	21247031142	9	21250176012	2409221626	0.146	0.132	0.012	9.
☐	10	Row...	200	appli...	[	tron	trx	TRON	https://coin-i...	0.158	13708756876	10	13708773705	294328153	0.16	0.158	-0	-0

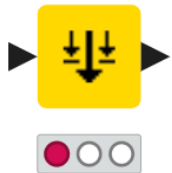
Filtered table (Table)

Rows: 10 | Columns: 7

☐	#	RowID	ids String	symbols String	names String	current_prices Number (double)	market_caps Number (long)	market_cap_r... Number (integer)	last_updateds String
☐	1	Row...	bitcoin	btc	Bitcoin	68,445	1353894327828	1	2024-10-19T03:5...
☐	2	Row...	ethereum	eth	Ethereum	2,646.49	318630708556	2	2024-10-19T03:5...
☐	3	Row...	tether	usdt	Tether	1	119981028809	3	2024-10-19T03:5...
☐	4	Row...	binancecoin	bnb	BNB	598.83	87399219145	4	2024-10-19T03:5...
☐	5	Row...	solana	sol	Solana	154.62	72676364138	5	2024-10-19T03:5...
☐	6	Row...	usd-coin	usdc	USDC	1	34937218632	6	2024-10-19T03:5...
☐	7	Row...	ripple	xrp	XRP	0.548	31051653136	7	2024-10-19T03:5...
☐	8	Row...	staked-ether	steth	Lido Staked Ether	2,646.29	25850084860	8	2024-10-19T03:5...
☐	9	Row...	dogecoin	doge	Dogecoin	0.145	21247031142	9	2024-10-19T03:5...
☐	10	Row...	tron	trx	TRON	0.158	13708756876	10	2024-10-19T03:5...

Column Filter

Column Filter



Dialog - 5:4 - Column Filter

Column filter

Manual Wildcard Regex Type

Search Aa

Excludes

- Status
- Content type
- body
- images
- fully_diluted_valuations
- total_volumes
- high_24hs

Includes

- ids
- symbols
- names
- current_prices
- market_caps
- market_cap_ranks
- Any unknown column

Cancel Ok and Execute

เลือก Column ที่ต้องการ

Row Filter

Filtered table (Table)

Rows: 10 | Columns: 7

☐	#	RowID	ids String	symbols String	names String	current_prices Number (double)	market_caps Number (long)	market_cap_r... Number (integer)	last_updateds String
☐	1	Row...	bitcoin	btc	Bitcoin	68,445	1353894327828	1	2024-10-19T03:5...
☐	2	Row...	ethereum	eth	Ethereum	2,646.49	318630708556	2	2024-10-19T03:5...
☐	3	Row...	tether	usdt	Tether	1	119981028809	3	2024-10-19T03:5...
☐	4	Row...	binancecoin	bnb	BNB	598.83	87399219145	4	2024-10-19T03:5...
☐	5	Row...	solana	sol	Solana	154.62	72676364138	5	2024-10-19T03:5...
☐	6	Row...	usd-coin	usdc	USDC	1	34937218632	6	2024-10-19T03:5...
☐	7	Row...	ripple	xrp	XRP	0.548	31051653136	7	2024-10-19T03:5...
☐	8	Row...	staked-ether	steth	Lido Staked Ether	2,646.29	25850084860	8	2024-10-19T03:5...
☐	9	Row...	dogecoin	doge	Dogecoin	0.145	21247031142	9	2024-10-19T03:5...
☐	10	Row...	tron	trx	TRON	0.158	13708756876	10	2024-10-19T03:5...

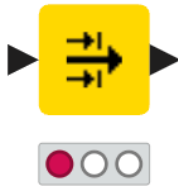
Included Rows (Table)

Rows: 5 | Columns: 7

☐	#	RowID	ids String	symbols String	names String	current_prices Number (double)	market_caps Number (long)	market_cap_r... Number (integer)	last_updateds String
☐	1	Row...	bitcoin	btc	Bitcoin	68,445	1353894327828	1	2024-10-19T03:5...
☐	2	Row...	ethereum	eth	Ethereum	2,646.49	318630708556	2	2024-10-19T03:5...
☐	3	Row...	tether	usdt	Tether	1	119981028809	3	2024-10-19T03:5...
☐	4	Row...	binancecoin	bnb	BNB	598.83	87399219145	4	2024-10-19T03:5...
☐	5	Row...	solana	sol	Solana	154.62	72676364138	5	2024-10-19T03:5...

Row Filter

Row Filter



Dialog - 5:6 - Row Filter

Filter

Criterion 1

Filter column: market_cap_ranks

Operator: Less than or equal

Value: 5

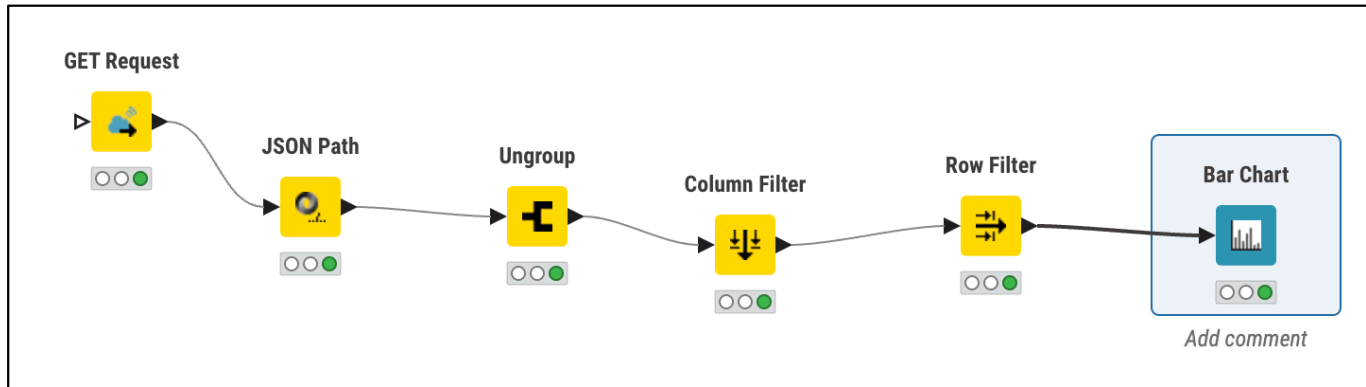
+ Add criterion

Output เพิ่มเงื่อนไขในการ Filter

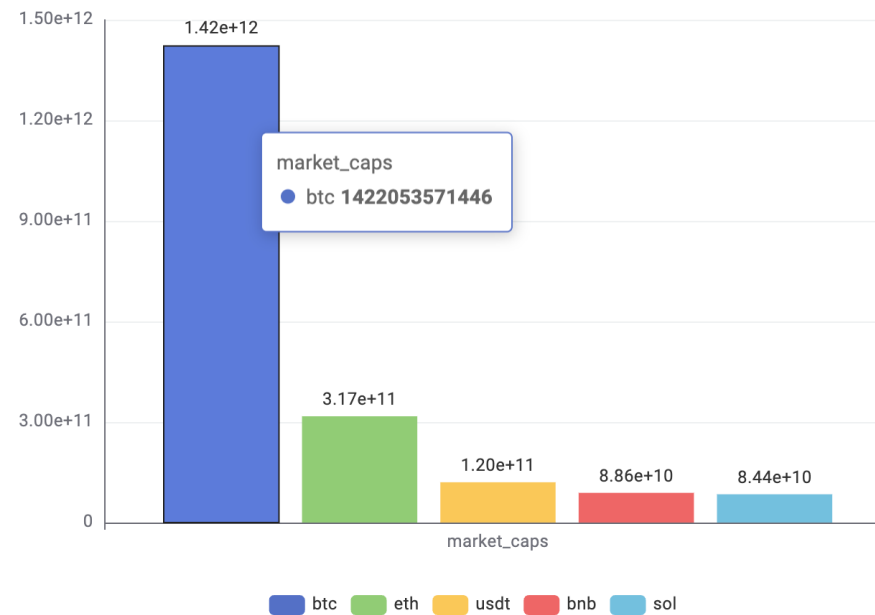
Column domains: Retain Compute

Filter behavior: Output matching rows Output non-matching rows

Cancel Ok and Execute



Top 5 Market Caps



Q&A

LAB: API Open Data

- สำรวจ Open API บน data.go.th



LAB: API

- สมัครใช้งาน <https://data.go.th/> เพื่อรับ USER TOKEN สำหรับเป็น API Key



ชุดข้อมูล กลุ่มชุดข้อมูล

รายงาน COVID-19 ประจำวัน ข้อมูลประจำประเทศไทย

องค์กร : กรมควบคุมโรค

ชุดข้อมูลที่มีคุณค่าสูง

รายงานผู้ยืนยันประจำวัน จาก กรมควบคุมโรค

สามารถติดตามสถานการณ์ได้ที่ <https://ddc.moph.go.th/viralpneumonia> และ API : <https://covid19.ddc.moph.go.th/>

ข้อมูลผู้ติดเชื้อรายเคสจะมีจำนวนไม่เท่ากับยอดรายงานผลรวมในปัจจุบัน เพราะกรมควบคุมโรคต้องดำเนินการตรวจสอบข้อมูลในรหัสเอกสารผู้ติดเชื้อรายเคสก่อนการเผยแพร่ ซึ่งการเผยแพร่ API ของ สพร หรือ API ของกรมควบคุมโรค จะเป็นยอดจำนวนผู้ติดเชื้อที่น้อยกว่ายอดรวมปัจจุบัน "รายงาน COVID-19 ประจำวัน ข้อมูลประจำประเทศไทย"

CKAN Data API

ดูรายละเอียดเพิ่มเติมใน เอกสารคู่มือการใช้งาน Data.go.th API

กำลังเรียกข้อมูล »

ตัวอย่างเรียกข้อมูล (5 ผลลัพธ์แรก)

https://open.data.go.th/get-ckan/datastore_search?resource_id=4688036a-cd89-45f7-a0c3-a34dbe35a50c&limit=5

ตัวอย่างเรียกข้อมูล (ผลลัพธ์มีคำว่า 'jones')

https://open.data.go.th/get-ckan/datastore_search?resource_id=4688036a-cd89-45f7-a0c3-a34dbe35a50c&q=jones

ZIP All resource data

235 downloads

รายงานจำนวนผู้ติดเชื้อ COVID-19 ประจำวันเริ่มตั้งแต่วันที่ 12 เมษายน 2565

7,222 downloads

รายงานจำนวนผู้ติดเชื้อ COVID-19 ประจำวันเริ่มตั้งแต่วันที่ 28...

4,320 downloads

รายงานจำนวนผู้ติดเชื้อ COVID-19 ประจำวันเริ่มตั้งแต่วันที่ 27 ตุลาคม 2564

5,899 downloads

มุมมอง

Data API

Visualization

ฝังตัว

แสดงผลเต็มหน้าจอ

ข้อมูลทรัพยากร : รายงานจำนวนผู้ติดเชื้อ COVID-19 ประจำวันเริ่มตั้งแต่วันที่ 12 เมษายน 2565

ดาวน์โหลด

วันที่ปรับปรุงไฟล์/ลิงก์ข้อมูลใน catalog 1 พฤษภาคม 2565

วันที่สร้างไฟล์/ลิงก์ข้อมูลใน catalog 12 เมษายน 2565

รูปแบบของไฟล์ชุดข้อมูล

application/vnd.openxmlformats-officedocument.spreadsheetml.sheet

รหัสทรัพยากรข้อมูล

4688036a-cd89-45f7-a0c3-a34dbe35a50c

วันที่เริ่มต้นสร้าง

2022-04-12

วันที่ปรับปรุงเนื้อหาข้อมูลล่าสุด

2022-04-12

ทดสอบเรียก REST API ด้วย โปรแกรม POSTMAN

`https://opend.data.go.th/get-ckan/datastore_search?resource_id=4688036a-cd89-45f7-a0c3-a34dbe35a50c&limit=5`

GET `https://opend.data.go.th/get-ckan/datastore_search?resource_id=4688036a-cd89-45f7-a0c3-a34dbe35a50c&limit=100&offset=100`

Params • **Authorization** • Headers (7) Body Pre-request Script Tests Settings

Type	API Key	Key	api-key
The authorization header will be automatically generated when you send the request. Learn more about authorization		Value	
		Add to	Header

Body Cookies (1) Headers (8) Test Results

Pretty Raw Preview Visualize JSON

```
1
2  "help": "https://data.go.th/api/3/action/help_show?name=datastore_search",
3  "success": true,
4  "result": {
5    "include_total": true,
6    "limit": 5,
7    "records_format": "objects",
8    "resource_id": "4688036a-cd89-45f7-a0c3-a34dbe35a50c",
9    "total_estimation_threshold": null,
10   "records": [
11     {
12       "_id": 1,
13       "No.": 3905873,
14       "announce_date": "2022-04-12T00:00:00",
15       "Notified date": "2022-04-11T00:00:00",
16       "sex": "หญิง",
17       "age": 54,
18       "Unit": "ปี",
19       "nationality": "Thailand",
20       "province_of_isolation": "เชียงใหม่",
21       "risk": "อยู่ระหว่างการสอบสวน",
22       "province_of_onset": "เชียงใหม่",
23       "district_of_onset": "จอมทอง"
24     },
25     {
26       "_id": 2,
27       "No.": 3905874,
28       "announce_date": "2022-04-12T00:00:00",
29       "Notified date": "2022-04-11T00:00:00",
30       "sex": "หญิง",
31       "age": 36,
32       "Unit": "ปี",
33       "nationality": "Thailand".
```

ทดสอบเรียก REST API ด้วย โปรแกรม POSTMAN

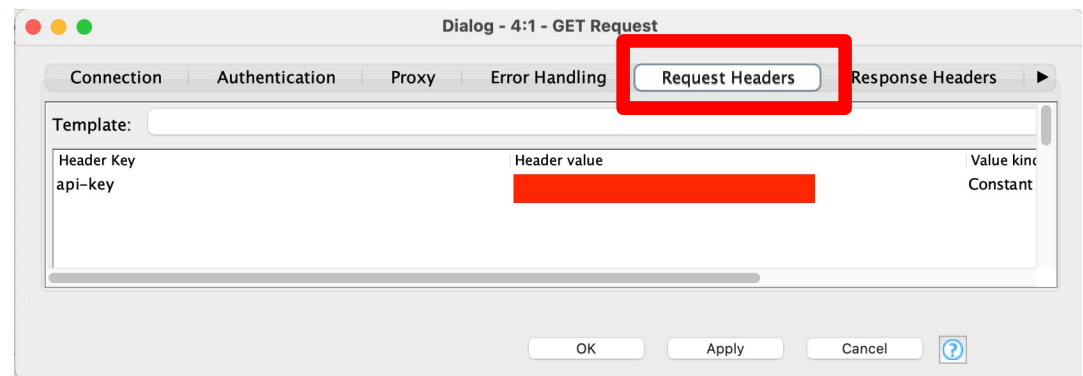
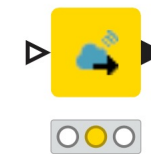
```
https://opend.data.go.th/get-ckan/datastore_search?  
resource_id=4688036a-cd89-45f7-a0c3-a34dbe35a50c  
&limit=100  
&offset=100
```

รันและสังเกตผลลัพธ์

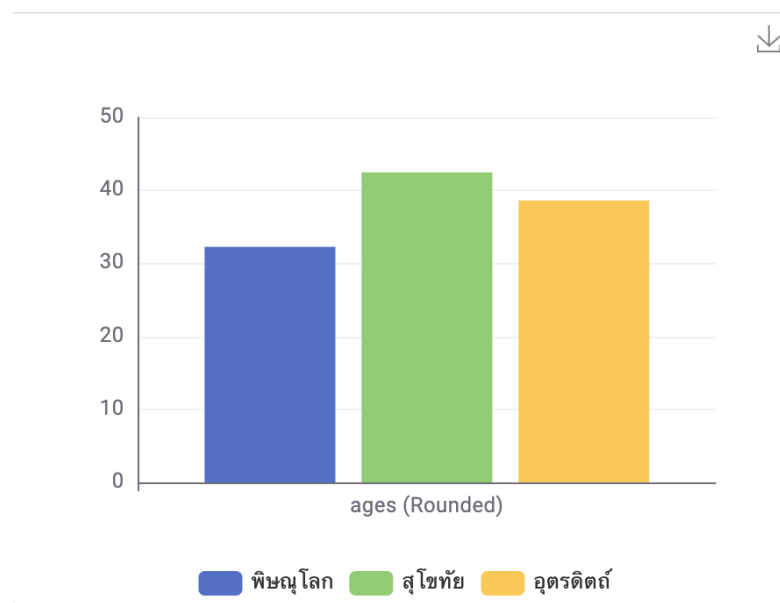
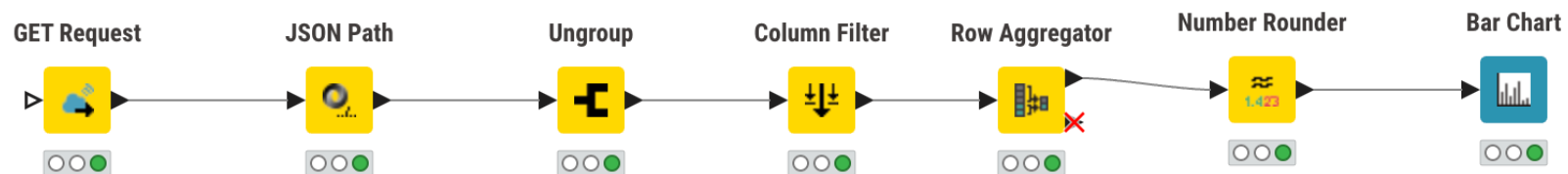
ทดสอบเรียก REST API ด้วย โปรแกรม KNIME

```
https://opend.data.go.th/get-ckan/datastore_search?  
resource_id=4688036a-cd89-45f7-a0c3-a34dbe35a50c  
&limit=100  
&offset=1000
```

GET Request

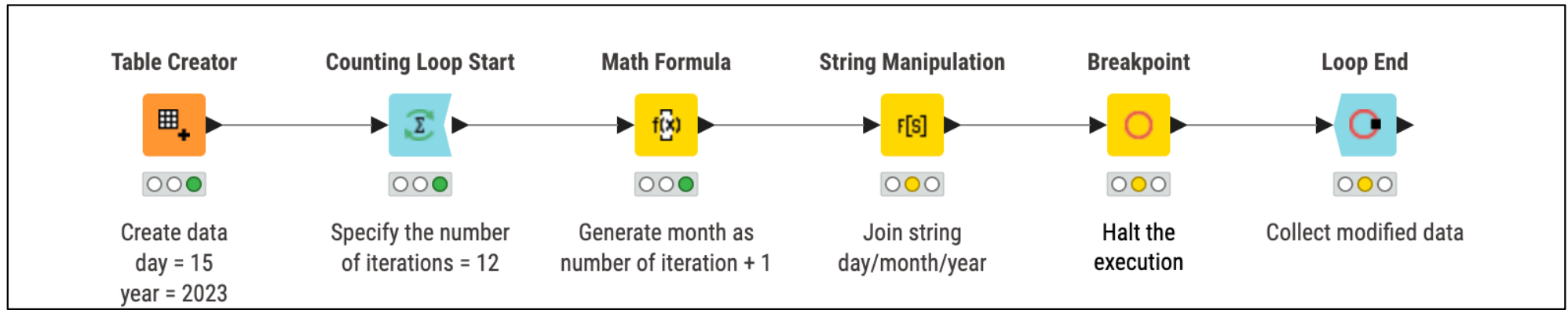


- รันและสังเกตผลลัพธ์
- Bar Chart อายุเฉลี่ยผู้ป่วยรายจังหวัด



Q&A

Use Case: Loop Functions



day Number (integer)	year String	month Number (double)	date String	Iteration Number (integer)
15	2023	1	15/1/2023	0
15	2023	2	15/2/2023	1
15	2023	3	15/3/2023	2
15	2023	4	15/4/2023	3
15	2023	5	15/5/2023	4
15	2023	6	15/6/2023	5
15	2023	7	15/7/2023	6
15	2023	8	15/8/2023	7
15	2023	9	15/9/2023	8
15	2023	10	15/10/2023	9
15	2023	11	15/11/2023	10
15	2023	12	15/12/2023	11

Ref: https://hub.knime.com/knime/spaces/Education/~ukW6Z9Nfqo_GYyKz/

Table Creator

Table Creator



Create data
day = 15
year = 2023

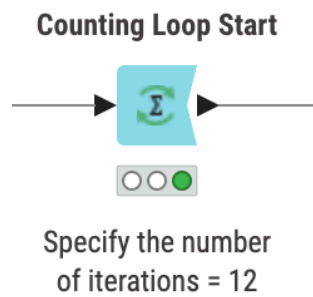
Dialog - 5:37 - Table Creator (Create data)

Table Creator Settings | Flow Variables | Job Manager Selection

Input line:

	I day	S year
Row0	15	2023
Row1		
Row2		
Row3		
Row4		
Row5		
Row6		
Row7		

Counting Loop



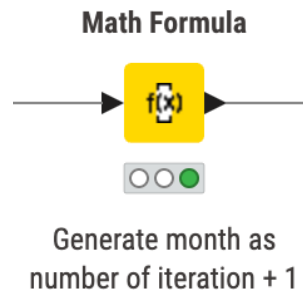
Dialog - 5:38 - Counting Loop Start (Specify the...

Number of loops 

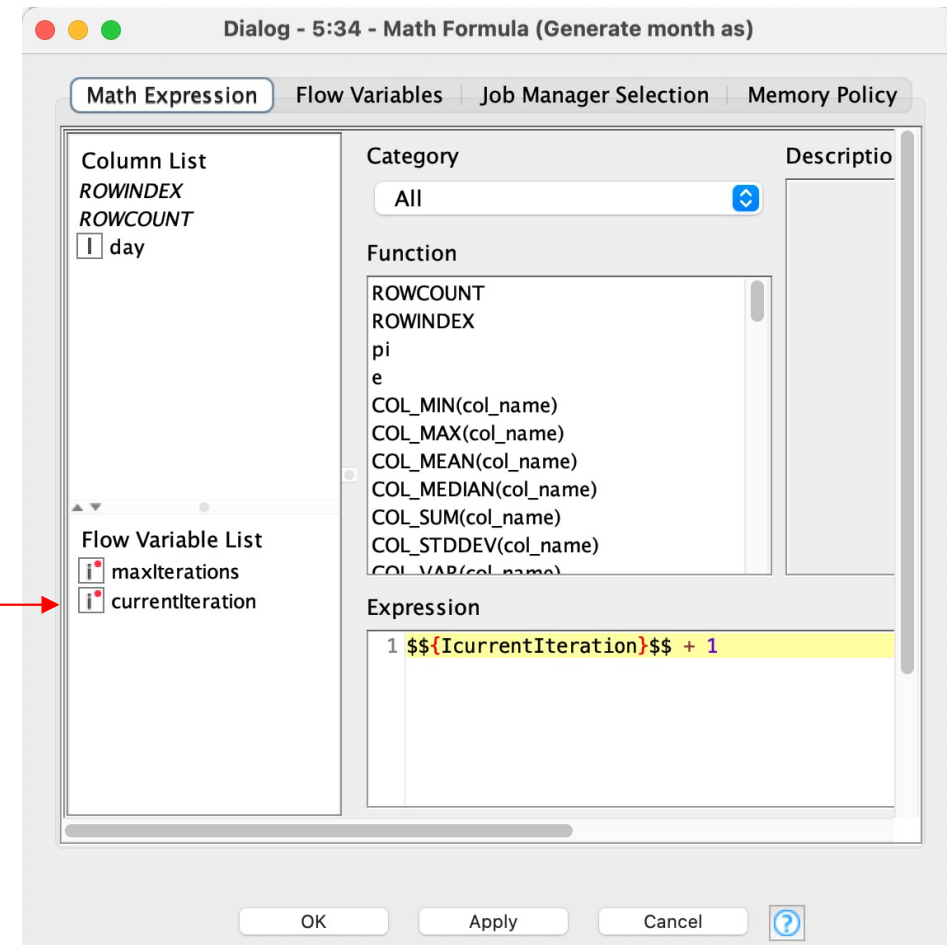
12  

Cancel Ok and Execute

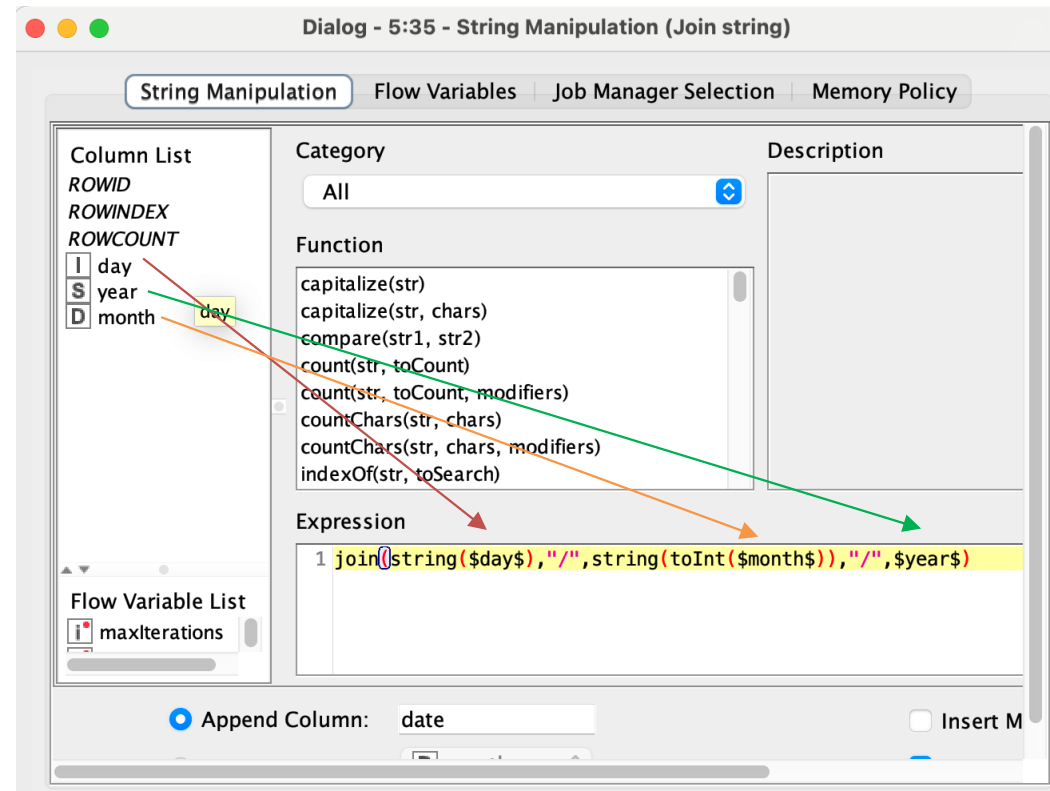
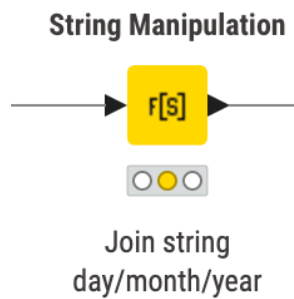
Math Formula



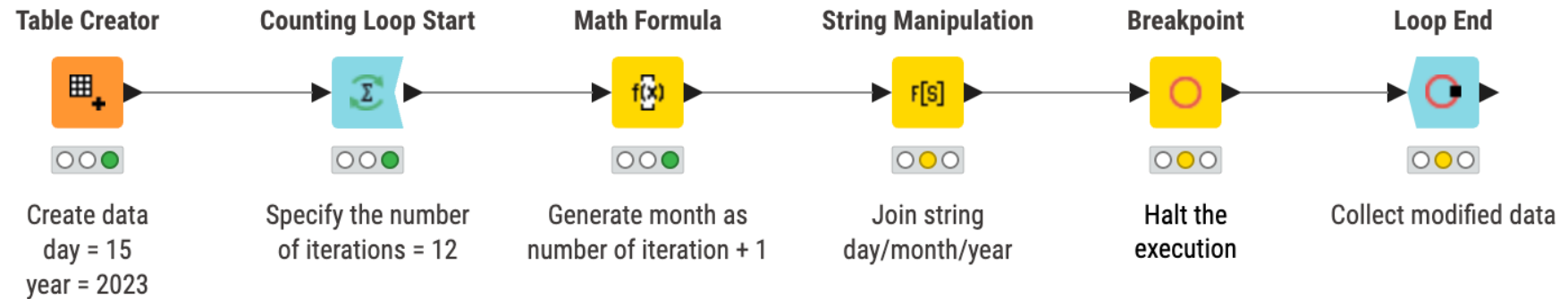
Double
Click !!



String Manipulation



Loop Functions

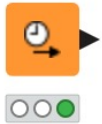


day Number (integer)	year String	month Number (double)	date String	Iteration Number (integer)
15	2023	1	15/1/2023	0
15	2023	2	15/2/2023	1
15	2023	3	15/3/2023	2
15	2023	4	15/4/2023	3
15	2023	5	15/5/2023	4
15	2023	6	15/6/2023	5
15	2023	7	15/7/2023	6
15	2023	8	15/8/2023	7
15	2023	9	15/9/2023	8
15	2023	10	15/10/2023	9
15	2023	11	15/11/2023	10
15	2023	12	15/12/2023	11

Tip !!!

- Try Create Date&Time Range

Create Date&Time Range



#	RowID	Date ↓ Local Date
1	Row0	2023-01-15
2	Row1	2023-02-15
3	Row2	2023-03-15
4	Row3	2023-04-15
5	Row4	2023-05-15
6	Row5	2023-06-15
7	Row6	2023-07-15
8	Row7	2023-08-15
9	Row8	2023-09-15
10	Row9	2023-10-15
11	Row10	2023-11-15
12	Row11	2023-12-15

Dialog - 6:47 - Create Date&Time Range

Options | Flow Variables | Job Manager Selection | Memory Policy

Output Settings

Output type: Date

New column name: Date

Mode Selection

Number of rows: ☐ Fixed: 12 ☒ Variable

Starting Point

Start: Date: 2023-01-15 Time: 16:21:15

Time Zone: Asia/Bangkok

☐ Use execution date&time

Ending Point

Interval: 1M

End: Date: 2023-12-15 Time: 16:21:15

☐ Use execution date&time

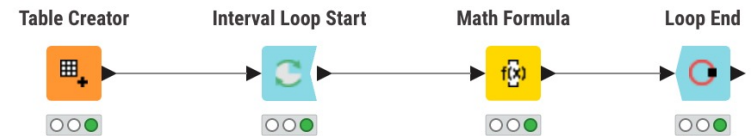
OK Apply Cancel ?

LAB: Loop Functions

- ใช้ Interval Loop Start เพื่อเรียก REST API หลายรอบ
- เปลี่ยนตัวแปรใน URL แต่ละรอบ โดยค่า offset 0, 100, 200, ..., 1000 โดยทำการ Get Request ครั้งละ 100 records

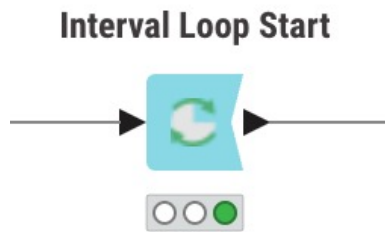
```
https://opend.data.go.th/get-ckan/datastore_search?  
resource_id=4688036a-cd89-45f7-a0c3-a34dbe35a50c  
&limit=100  
&offset=XX
```

ทดลอง flow นี้ก่อน



RowID	offset Number (double)	Iteration Number (integer)
Row0#0	0	0
Row0#1	100	1
Row0#2	200	2
Row0#3	300	3
Row0#4	400	4
Row0#5	500	5
Row0#6	600	6
Row0#7	700	7
Row0#8	800	8
Row0#9	900	9
Row0#10	1,000	10

Interval Loop



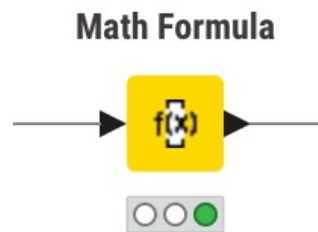
Dialog - 3:6 - Interval Loop Start

Standard settings | Flow Variables | Job Manager Selection ▶

From	0.0
To	1000.0
Step	100.0
Loop variable is	☐ double ☐ long ☒ integer
Variable prefix	loop_

OK Apply Cancel ?

Math Formular



Dialog - 6:34 - Math Formula (Generate month as)

Math Expression Flow Variables Job Manager Selection Memory Policy

Column List	Category	Description
ROWINDEX	All	
ROWCOUNT		
1 day		

Function

- ROWCOUNT
- ROWINDEX
- pi
- e
- COL_MIN(col_name)
- COL_MAX(col_name)
- COL_MEAN(col_name)
- COL_MEDIAN(col_name)
- COL_SUM(col_name)
- COL_STDDEV(col_name)
- COL_VAR(col_name)
- ln(x)

Flow Variable ...

- maxIterations
- currentIteration

Expression

```
1 ${currentIteration} + 1
```

☒ Append Column: month

☐ Replace Column: S year

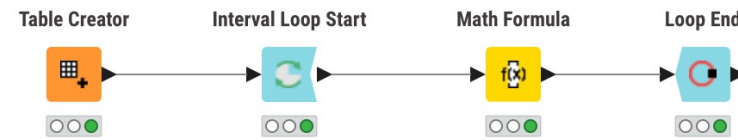
☐ Convert to Int

OK Apply Cancel ?

ส่งค่าออกเป็น
จำนวนเต็ม

เพิ่ม column ใหม่

แทนค่าใน
column เดิม



↓

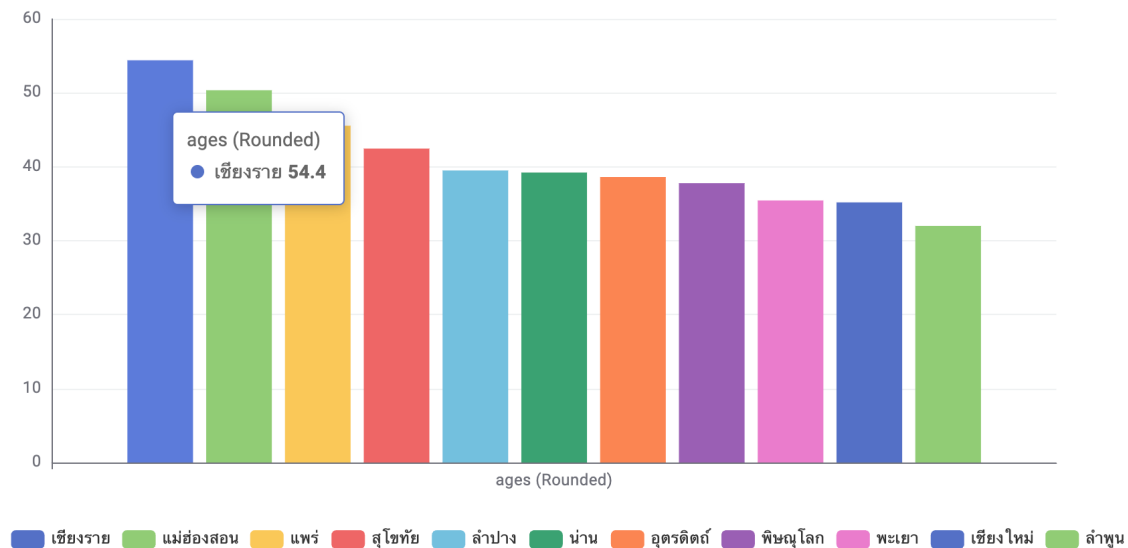
RowID	offset <i>Number (double)</i>	Iteration <i>Number (integer)</i>
Row0#0	0	0
Row0#1	100	1
Row0#2	200	2
Row0#3	300	3
Row0#4	400	4
Row0#5	500	5
Row0#6	600	6
Row0#7	700	7
Row0#8	800	8
Row0#9	900	9
Row0#10	1,000	10

LAB: Loop Functions

- เปลี่ยนตัวแปรใน URL แต่ละรอบ โดยค่า offset 0, 100, 200, ..., 1000 โดยทำการ Get Request ครั้งละ 100 records

```
https://opend.data.go.th/get-ckan/datastore_search?  
resource_id=4688036a-cd89-45f7-a0c3-a34dbe35a50c  
&limit=100  
&offset=XX
```

เป้าหมาย



Q&A